

Interpretable Control for Unstable Systems via SINDy-RL

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1 Introduction

Safety-critical autonomous systems increasingly require controllers that can be audited, formally verified, and deployed on embedded hardware — requirements that a nine-thousand-parameter neural network cannot satisfy [8, 9]. A closed-form polynomial controller, by contrast, fits in kilobytes, admits analytic stability arguments, and exposes every term to inspection. Sparse Identification of Nonlinear Dynamics (SINDy [2]) produces exactly such expressions by regressing state-transition data against a polynomial library and zeroing out negligible terms. The difficulty is that unstable systems cannot generate the near-equilibrium data SINDy needs without a stabilizing controller that does not yet exist. Zolman et al. [1] resolve this chicken-and-egg problem with SINDy-RL: a Dyna-style [6] loop that co-trains an E-SINDy surrogate and a PPO policy so each iteration improves both. We implement Algorithm 1 from Zolman et al. on the inverted double pendulum (IDP), resolving several non-obvious engineering obstacles to achieve convergence, and deliver (1) a data-efficient NN policy using $14.5\times$ fewer real-environment steps than a full-order baseline, and (2) a degree-3 polynomial distilled from that policy.

2 Technical Background

2.1 The Testbed

The `InvertedDoublePendulum-v5` environment (MuJoCo/Gymnasium) has two rigid links of equal length $L_1 = L_2 = 0.6$ m on a sliding cart. The 6-dim physical state is $\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$; the 9-dim observation replaces raw angles with sin/cos encodings. The single action is cart force. Tip height $h = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$ falls at 1.0 m, leaving a 0.2 m near-upright band. The per-step reward is $r_k = 10 \cdot \mathbf{1} - (h_k - 2)^2 - 0.01 x_{\text{tip}}^2 - \varepsilon \|\dot{\theta}\|^2$, with episodes capped at 1,000 steps (50 s).

2.2 SINDy-C: Sparse Dynamics Identification with Control

SINDy [2] identifies discrete-time dynamics by regressing the state increment against a polynomial library:

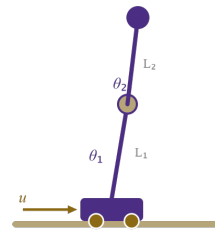


Figure 1. IDP geometry. State $\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$. Tip height $h \in [0, 1.2]$ m; episode ends at $h \leq 1.0$ m.

$$\mathbf{x}_{k+1} - \mathbf{x}_k = \underbrace{\Theta(\mathbf{x}_k, u_k)}_{\text{library}} \cdot \underbrace{\Xi}_{\text{sparse coefficients}}$$

For control-affine systems (SINDy-C [3]), the input u_k enters the library directly. The Sequentially Thresholded Least Squares (STLSQ) algorithm zeros coefficients below threshold λ , promoting sparsity in Ξ . A degree- d library over n variables contains $\binom{n+d}{d}$ terms; for the IDP’s 7-dim state-action vector, degree-2 gives 36 features and degree-3 gives 120 — a distinction that proved critical (§4.3).

2.3 E-SINDy: Ensemble Uncertainty Quantification

Fitting $M=10$ independent SINDy models on 80% bootstrap subsamples yields a free uncertainty estimate [4]: at each step, `predict(x, u)` returns $(\mu_\Delta, \sigma_\Delta)$ across ensemble members. High σ_Δ signals extrapolation beyond the training distribution. Following Zolman et al. §3.5 [1], we convert this into an active pessimistic penalty: surrogate reward is reduced by $\kappa \cdot \text{mean}(\sigma_\Delta)$ per step ($\kappa = 5.0$), steering PPO away from high-uncertainty states.

2.4 Dyna-Style MBRL and Behavioral Cloning

The Dyna architecture [6] alternates cheap model-based rollouts with real data collection. In SINDy-RL [1], the surrogate is the E-SINDy ensemble and the planner is PPO [5]. Figure 2 shows the control loop; in SINDy-RL the Environment box is instantiated twice — as the E-SINDy surrogate during training and as real MuJoCo during data collection.

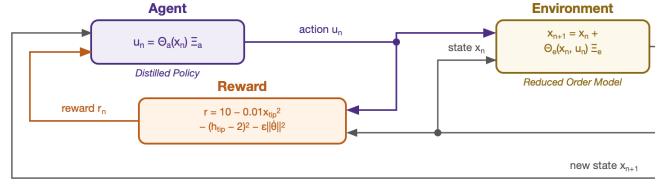


Figure 2. The RL control loop. In SINDy-RL, the Environment is instantiated as the E-SINDy surrogate for cheap policy training and as real MuJoCo for data collection and evaluation.

A Schroeder multi-sine sweep [10] bootstraps the initial dataset $\mathcal{D}$. Each Dyna iteration refits E-SINDy on near-upright transitions, runs PPO for 100k surrogate steps (warm-started from the prior policy), then collects 4,000 real transitions. After convergence, the best checkpoint is distilled via behavioral cloning:

$$\min_{\Xi} \|\Theta_{\text{obs}}(X) \Xi - U^*\|_2 \quad (\text{STLSQ}, \lambda = 0.05)$$

Perturbation augmentation [7] — adding Gaussian noise to expert states and re-querying the NN oracle — expands the 50k-transition dataset $5\times$ to mitigate distribution shift without additional simulator rollouts.

3 Methods

3.1 Baseline

A standard PPO agent ([64,64] MLP tanh, 9,731 params, 8 parallel envs) is trained for 400,000 real steps, serving as the performance ceiling only — it is not used as a distillation teacher.

3.2 SINDy-RL Pipeline

The pipeline in `notebooks/sindy-rl.ipynb` runs five stages: (1) Schroeder bootstrap — 300 episodes collecting 2,897 transitions; (2) E-SINDy fit — filter to near-upright states ($h > 1.10$ m), fit 10 degree-3 SINDy-C models on 80% subsamples, stack into a `FastEnsemblePredictor`; (3) surrogate PPO — 100k steps inside `EnsembleSurrogateEnv` with uncertainty penalty, early-stopped if mean episode length stays below 5 after 50k steps; (4) real data collection — 4,000 MuJoCo steps appended to $\mathcal{D}$; (5) repeat with warm-start, rolling back to best checkpoint if exploitation is detected. Distillation follows the loop: 50k expert transitions augmented $5\times$ and fit with a degree-3 STLSQ polynomial on the 8-dim observation.

3.3 Metrics

We evaluate on real-environment step count (data efficiency), success rate (episodes lasting at least 500 steps), mean episode length, SINDy RMSE (surrogate quality), and distillation R^2 and term count (compactness).

4 Preliminary Results

4.1 Baseline

Full-order PPO achieves mean reward $9,324 \pm 2$, 100% success, and mean episode length 1,000/1,000 steps at a cost of 400,000 real simulator interactions and a 9,731-parameter opaque network. This is the performance ceiling.

4.2 Dyna Loop Convergence

The Dyna loop converged in four iterations using 27,512 real steps — **$14.5\times$ fewer than the baseline**. Figure 3 shows episode length and surrogate RMSE across iterations; the RMSE rise at iterations 3–4 reflects the improving policy visiting states further from vertical, yet the surrogate remained accurate enough in the near-upright band to train a transferable policy. A post-loop evaluation of the iteration-4 checkpoint gave **75% success and mean episode length 763 steps**.

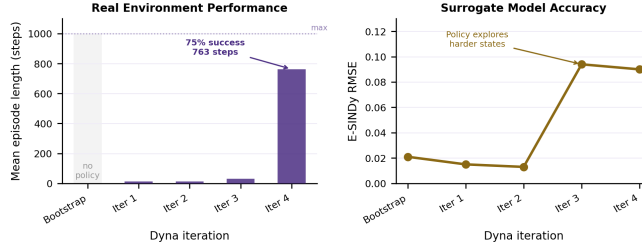


Figure 3. Dyna loop convergence. Bars (left): mean real episode length. Line (right): SINDy RMSE. Iteration 4 reached 805 mean real steps and 80% success during the loop; a post-loop evaluation of the selected checkpoint reached 763 mean steps and 75% success.

4.3 What Didn’t Work: Obstacles

Three obstacles prevented convergence on early attempts. First, a **degree-2 RMSE ceiling**: over 25 iterations the RMSE oscillated at 0.10–0.16 regardless of data volume (5k→90k transitions), because a 36-feature degree-2 library cannot express the inter-mode coupling that dominates IDP dynamics. Fix: `SINDY_DEGREE=3` (120 features), which dropped RMSE to 0.013 within two iterations. Second, a **near-upright filter geometry bug**: the filter threshold was set to 1.6 m by inheriting a reward-shaping constant (`TIP_HEIGHT_TARGET=2.0`) that has no relation to the pendulum’s physical reach of $L_1 + L_2 = 1.2$ m, silently making the filter a no-op every iteration. Fix: `SINDY_H_MIN=1.10` m, derived from segment geometry. Third, **surrogate exploitation**: in a diagnostic run, surrogate reward jumped 9× while real episode length collapsed 87%, as the policy found action sequences the polynomial predicted as highly rewarding but which were physically invalid. Neither uncertainty penalization nor rollback alone is sufficient — shared polynomial basis means all ensemble members extrapolate identically, so disagreement-based penalties are blind to the exploit; rollback limits damage but cannot prevent it. Fix: both mechanisms together.

A parallel Track-A diagnostic reinforced this failure mode. In a simpler fixed-surrogate variant, PPO was trained inside a SINDy model fit from near-upright MuJoCo probes and then transferred directly to real MuJoCo. The best polynomial SINDy-PPO checkpoint reached only 31.2 mean real steps with 0% success, and a trig-library branch reached only 17.8 mean real steps with 0% success despite improved one-step diagnostics. This suggests that PPO transfer needs surrogate accuracy over long recursive rollouts and over the policy-induced state-action distribution, not just low one-step error near upright.

4.4 Policy Distillation

Behavioral cloning from the best Dyna checkpoint produced a 165-term degree-3 polynomial achieving **65% success and mean episode length 672** (Figure 4). Three implementation issues required resolution: the distillation teacher

must be the best-checkpoint policy, not the final loop policy (which may have drifted during surrogate training); degree-2 gave $R^2 \approx 0.905$ regardless of data, requiring degree-3 to reach $R^2 = 0.9916$; and perturbation augmentation was needed to close the distribution-shift gap between the NN’s natural trajectory distribution and deployment states.

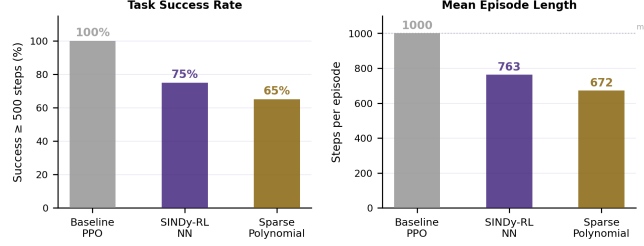


Figure 4. Method comparison. Success rate and mean episode length for baseline PPO, SINDy-RL NN, and the distilled sparse polynomial.

STLSQ retained all 165/165 policy terms, so the polynomial is compact but not sparse. It is still $59\times$ smaller than the 9,731-param NN and exposes recognizable dominant terms: bias, proportional angle feedback, velocity damping, and inter-pole coupling. The remaining small cubic cross-terms appear to encode NN residuals rather than clean physical structure. The clearest result is data efficiency: 27,512 Dyna steps ($14.5\times$) for the NN and 77,512 total steps ($5.2\times$) for the distilled controller, with distillation reducing success from 75% to 65% in exchange for a closed-form policy.

4.5 Summary, Code, and Next Steps

Compared with baseline PPO’s 400,000 real steps for 100% success, SINDy-RL used 27,512 real Dyna steps for 75% success and the distilled 165-term polynomial used 77,512 total real steps for 65% success. Next steps are STLSQ threshold ablations and direct training in a structure-constrained surrogate. Code: https://github.com/falconej1/ME_595; key notebooks are `full-order-simulation.ipynb` and `sindy-rl.ipynb`; Professor Michelle Hickner added as collaborator.

CRedit and AI-Assisted Work Disclosure. Patrick Smith led SINDy-RL implementation, baseline PPO, distillation, and obstacle diagnosis. Andrew Falcone supported methodology, diagnostic experiments, visualization, and writing/review. Both authors contributed to interpretation. AI tools assisted with notebook organization, code drafting, debugging, and experiment iteration; the authors reviewed and executed the work and are responsible for the submitted analysis.

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